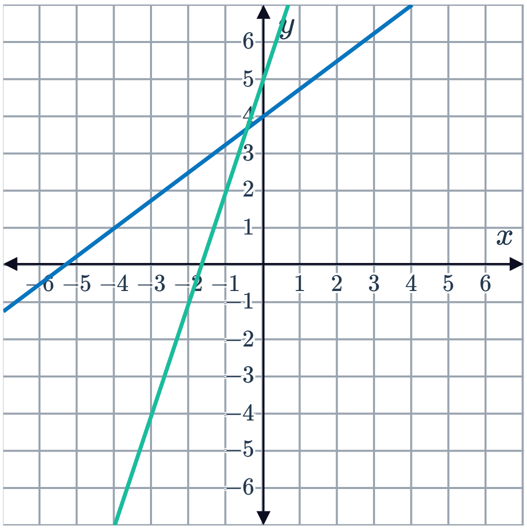
Systems of linear equations

1 State the number of solutions for each of the following graphed system of two linear equations:

a



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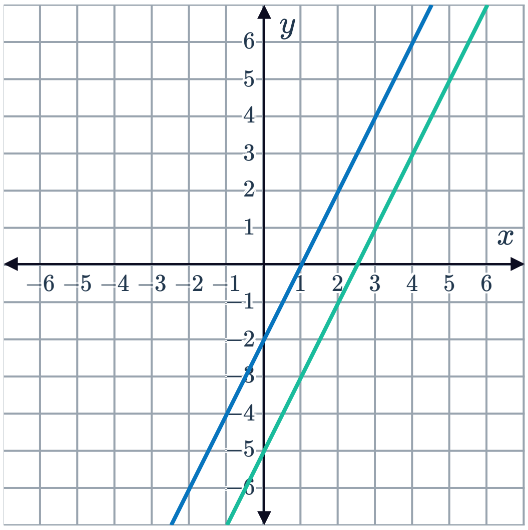

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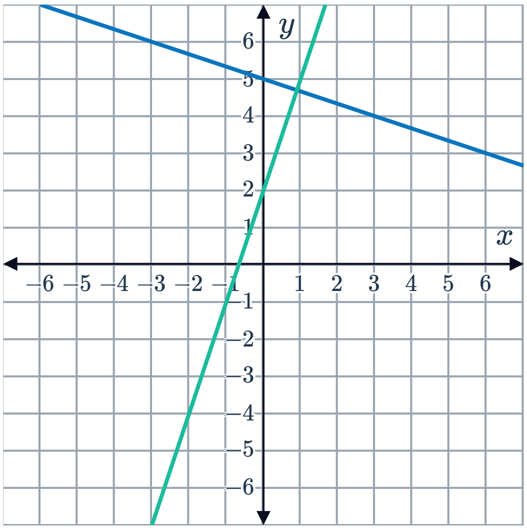
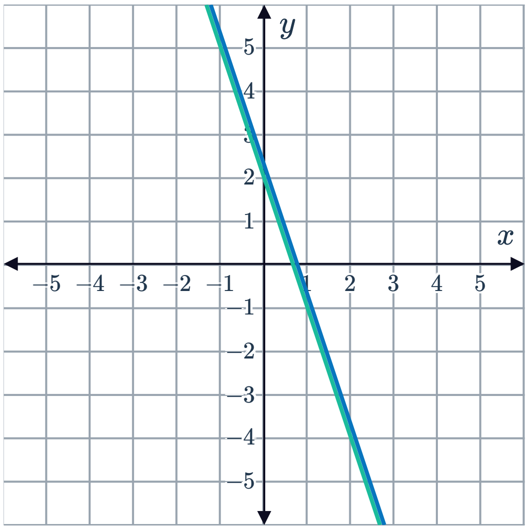


d



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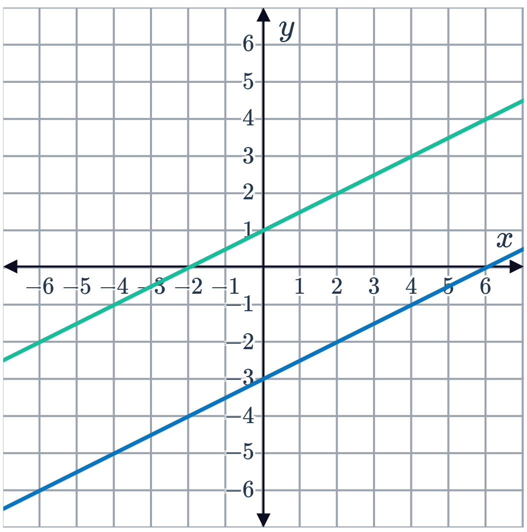
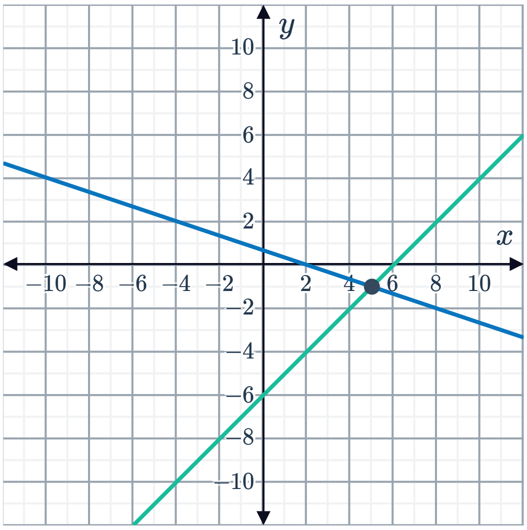


2 Consider the following graphs of systems [redacted]ear equations:

i How many solutions does this system of equations have?

ii Write down the solution(s) to the system of equations.

a



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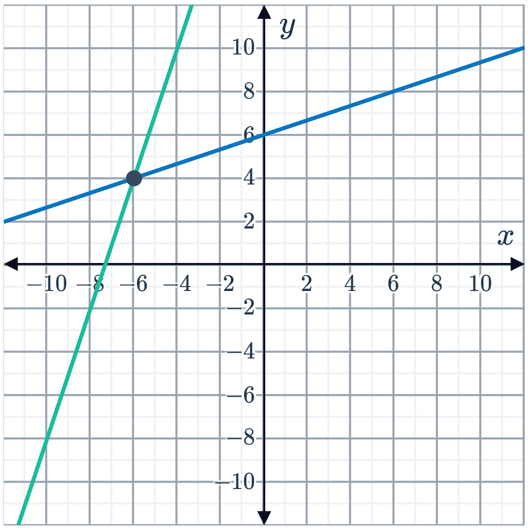

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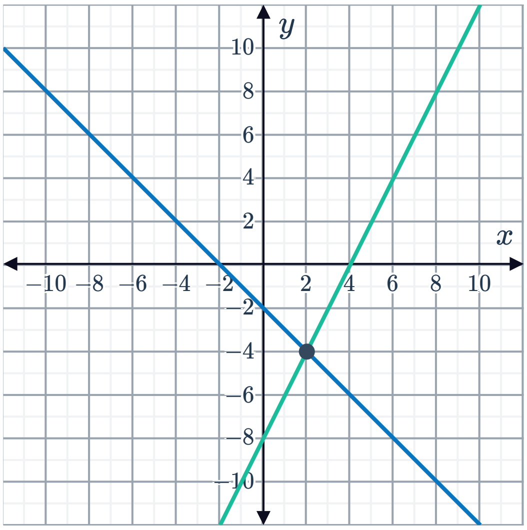

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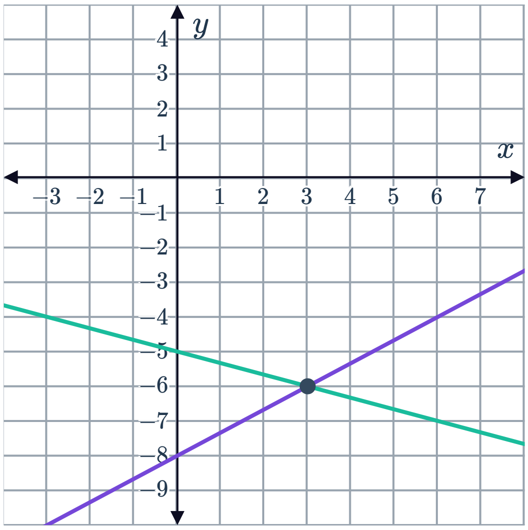

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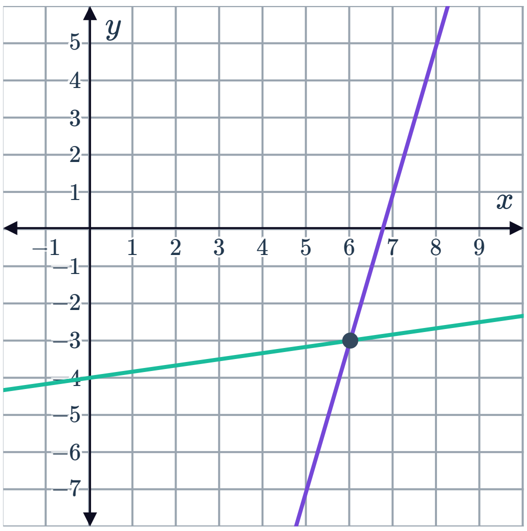

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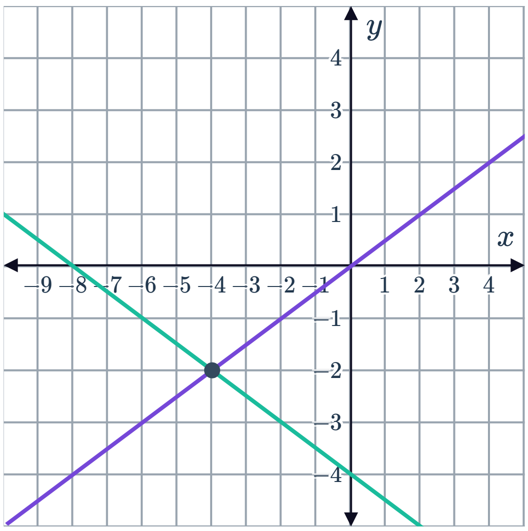
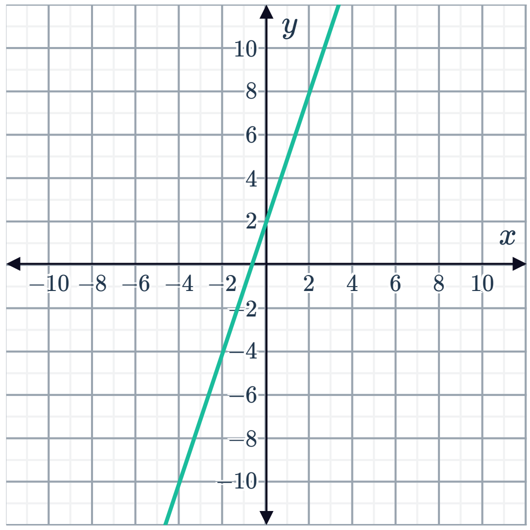


3 Consider the graph of the equation $y = 3x + 2$:

If a second line intersects this line at the point $(0, 2)$, can we determine the equation of the second line? Explain your answer.

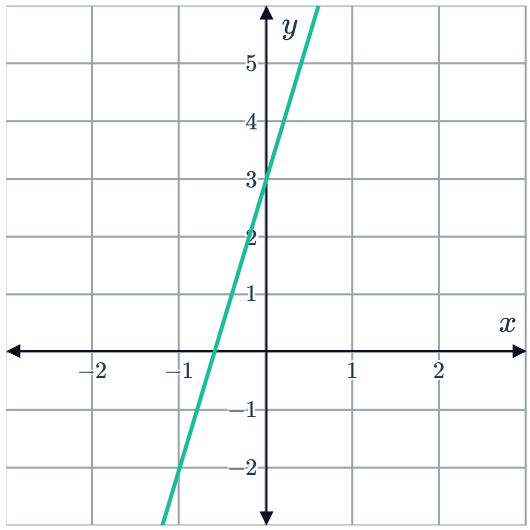


4 Consider the graph of the equation $y = 5x + 3$:

A second line, $y = mx + b$, intersects this line at the one point $(0, 3)$.

a State the value of b. Explain your answer.

b State the value that m cannot be equal to. Explain your answer.



5 A system of linear equations has no solutions. One of the equations of the system is $y = -3x - 2$. Determine whether the following could be the other equation of the system.

a $y = -3x - 3$

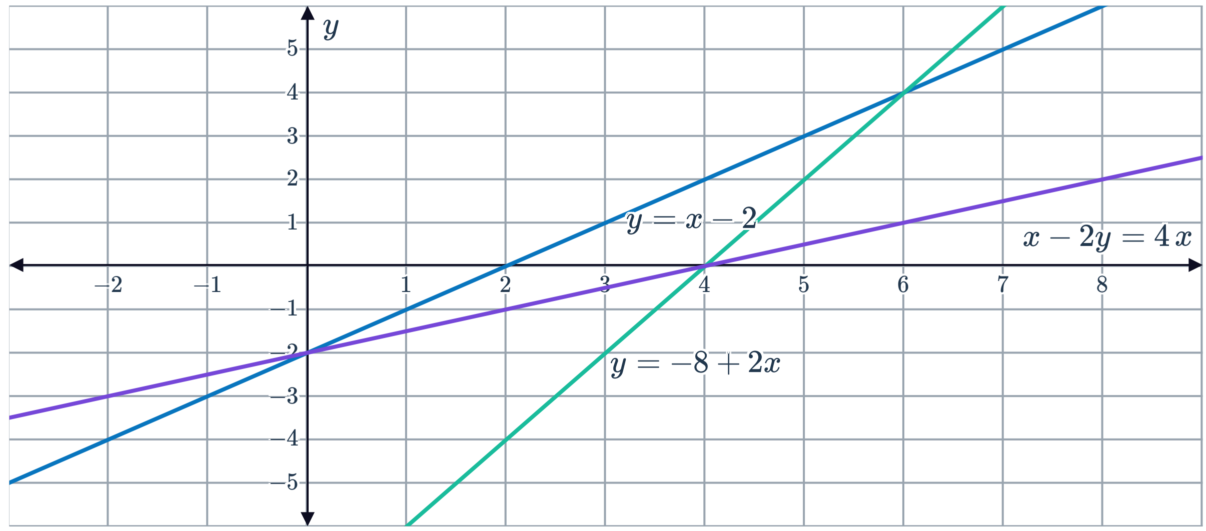
b $y = \frac{x}{3} - 3$

c $y = -\frac{x}{3} - 2$

d $y = 3x + 2$

Solve systems graphically

6 Use the given graph to solve each pair of simultaneous equations:



a $y = x - 2$
$y = -8 + 2x$

b $y = x - 2$
$x - 2y = 4$

c $y = -8 + 2x$
$x - 2y = 4$

7 For each pair of linear equations:

i Sketch the lines on the same number plane.

ii State the coordinates which satisfies both equations.

a $L_1 : y = 4x - 4$
$L_2 : y = 8 - 2x$

b $L_1 : y = x + 9$
$L_2 : y = -x - 9$

c $L_1 : y = x + 3$
$L_2 : y = -x + 3$

d $L_1 : 2x - 4y = -8$
$L_2 : -4x + 2y = -8$

e $L_1 : y = 3x + 6$
$L_2 : y = -x + 2$

f $L_1 : y = -x - 3$
$L_2 : y = -1$

g $L_1 : y = 4x - 3$
$L_2 : y = 4 - 3x$

8 Consider the following linear equations:

$$L_1 : y = -2x - 5$$
$$L_2 : y = -2x + 4$$

a Sketch L_1 and L_2 on the same number plane.

b Is there a coordinate that satisfies the two equations simultaneously? Explain your answer.

Applications

9 A rectangular zone is to be 3 m longer than it is wide, with a total perimeter of 18 m.

- **a** Let y represent the length of the rectangle and x represent the width. Construct two equations that represent this information.
- **b** Sketch the two lines on the same number plane.
- **c** Hence, find the length and width of the rectangle.

10 A band plans to record a demo at a local studio. The cost of renting studio A is $\$250$ plus $\$50$ per hour. The cost of renting studio B is $\$50$ plus $\$100$ per hour. The cost, y, in dollars of renting the studios for x hours can be modelled by the linear system:

- Studio A: $y = 50x + 250$
- Studio B: $y = 100x + 50$

- **a** Sketch the two lines on the same number plane.
- **b** State the coordinate which satisfies both equations.
- **c** What does the coordinate from part (d) mean?

11 Michael plans to start taking an aerobics class. Non-members pay $\$4$ per class. Members pay a $\$10$ one-time fee, but only have to pay $\$2$ per class. The monthly cost, y, of taking x classes can be modelled by the linear system:

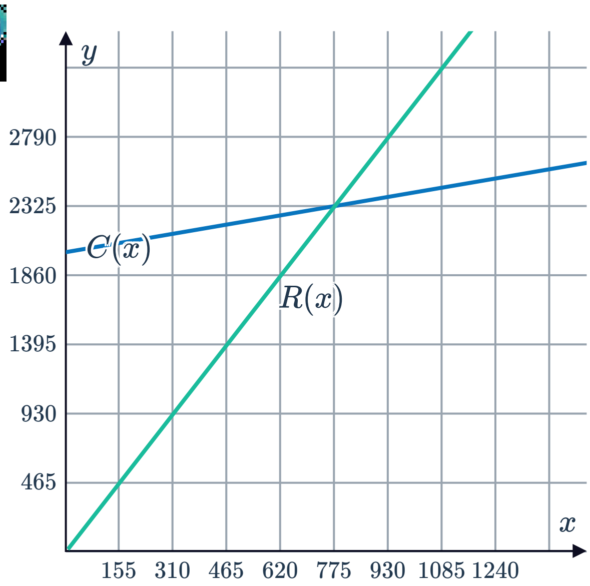
- Non-members: $y = 4x$
- Members: $y = 2x + 10$

- **a** Sketch the two lines on the same number plane.
- **b** State the coordinate which satisfies both equations.
- **c** What does the coordinate from part (b) mean?

12 The cost of manufacturing toys, C, is related to the number of toys produced, n, by the formula $C = 400 + 2n$. The revenue, R, made from selling n toys is given by $R = 4n$.

- **a** Sketch the graphs of cost and revenue on the same number plane.
- **b** How many toys need to be produced for the revenue to equal the cost?
- **c** State the meaning of the y-coordinate of the point of intersection.

13 Given the cost function $C(x) = 0.4x + 20$[redacted] and the revenue function $R(x) = 3x$, find the coordinates of the point of intersection, or the break-even point.



14 The two equations $y = 3x + 35$ and $y = 4x$ represent Laura's living expenses and income from work respectively.

- **a** Find the point of intersection of the two equations.
- **b** Sketch both equations on the same number plane.
- **c** State the meaning of the point of intersection of the two lines.

15 The two equations $y = 4x + 400$ and $y = 6x$ represent a company's revenue and expenditure respectively.

- **a** Find the point of intersection of the two equations.
- **b** Sketch both equations on the same number plane.
- **c** State the meaning of the point of intersection of the two lines.